[INCH-POUND]
A-A-3041
October 24, 1996

COMMERCIAL ITEM DESCRIPTION

WRENCH, OPEN END, RATCHET (TAC PATTERN), FOR TUBE FITTINGS, ELECTRICAL CABLE TERMINALS, AND STUFFING TUBE GLAND NUTS

The General Services Administration has authorized the use of this Commercial Item Description for procurement of type I, class 1 wrench specified in GGG-W-646 for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description covers open-ratchet wrenches used for tube fitting, stuffing gland nuts, and applications where difficulties arise in the use of standard design sockets.

2. CLASSIFICATION.

2.1 Classification. This Commercial Item Description covers open ratchet, crowfoot wrench heads of the following styles and forms:

Style A - Multipurpose
Style B - Single purpose
Form A - Hexagon
Form B - Open end

3. SALIENT CHARACTERISTICS.

3.1 Design. The open ratchet, crowfoot wrench heads shall consist of a body, slotted rotating member, an enclosure plate for accessibility in repairs, and shall be reversible by inverting the wrench head. The wrenches shall be made of forged steel of such composition as to produce wrenches conforming to the physical requirements of this Commercial Item Description. The wrench shall have a hardness of between 38 to 52 on the Rockwell C scale. The style and form of the wrenches shall be similar to Figures 1, 2, or 3, as applicable. The dimensions shall be as specified in Tables I or II, as applicable, except style B, form B which shall be as specified in the contract or purchase order (see 7.2).

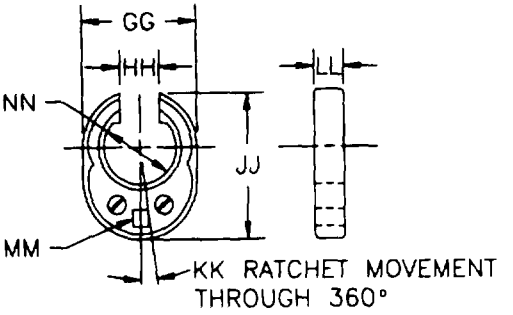


Figure 1. Style A, open ratchet, crowfoot, multipurpose wrench heads.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center, Kansas City, MO 64131.

Distribution Statement A. Approved for public release; distribution is unlimited.

FSC 5120

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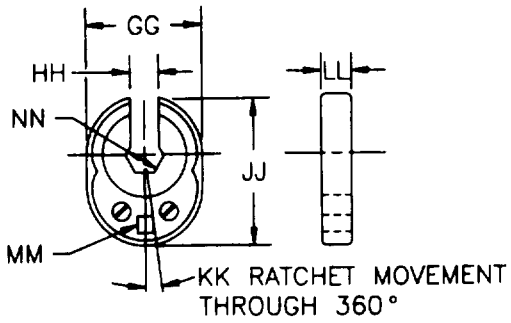


Figure 2. Style B, form A, open ratchet, crowfoot, single purpose, hexagon wrench heads.

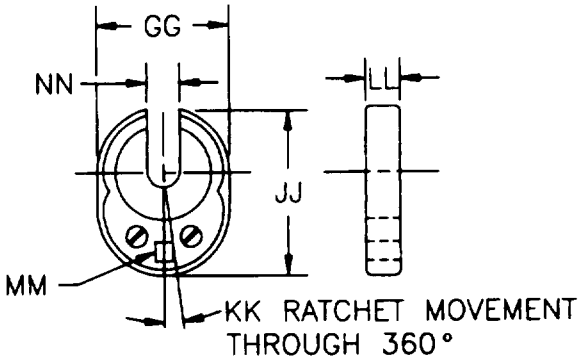


Figure 3. Style B, form B, open ratchets, crowfoot, single purpose, open end wrench heads.

TABLE I. Style A, open ratchet, crowfoot, multipurpose wrench heads

Component number	Rated torque (inch-pounds)	Code	Dimensions							
			GG 1/ (inches)	HH (Minimum)	JJ 1/ (Inches)	LL +.015 -.005 (Inch)	MM +.005 -.000 (Inch)	NN ±.005 (Inches)	KK	
									Degrees	Minute
2	1200	A-C	1.650	.645	2.210	.520	.381	.940	-	30
3	1500	D-I	2.150	.900	2.710	.644	.381	1.190	-	30
4	2250	J-M	2.750	1.400	3.470	.644	.508	1.720	-	30
5	3400	N-T	3.900	2.085	4.630	.710	.508	2.470	6	0
6	3750	V-Z	4.620	2.800	5.380	.770	.508	3.190	5	0

1/ Nominal dimension, held within drop-forging association tolerances

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TABLE II. Style B, form A, open ratchet, crowfoot, single purpose, hexagon wrench heads

Compo-nent number	Rated torque (Inch-pounds)	Hex nut size (Inches)	Dimensions								
			NN							KK	
			Minimum	Maximum	GG 1/ (Inches)	HH (Inches)	JJ 1/ (Inches)	LL + .015 - .005 (Inch)	MM + .005 - .000 (Inch)	Degrees	Minutes
10	225	9/16	.566	.575	1.630	.570 + .015 - .000	2.200	.322	.381	7	30
190	3000	2-11/16	2.701	2.722	4.645 ± 0.030	2.706 + .020 - .000	5.367	.520	.508	5	0

1/ Nominal dimension, held within drop-forging association tolerances.

3.2 Coatings. The wrench shall be coated with black oxide or black phosphate followed by a coating of rust preventive oil, or shall be electroplated with nickel/chromium. If electroplated with a nickel (Ni) coating, it shall be followed by a chromium (Cr) coating. There shall be no other coatings or undercoatings on the tool. The coating thickness requirements are as follows:

Ni = 0.000200 inch or greater
Cr = 0.000007 inch or greater

Within each surface area, one measurement of the thickness of each electroplated coating shall be conducted in accordance with any of the following methods, as appropriate:

ASTM B 487: Standard Method for Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section

ASTM B 499: Standard Test Method for Measurement of Coating Thickness by the Magnetic Method: Nonmagnetic Coatings of Magnetic Basis Metals

ASTM B 504: Standard Method for Measurement of Thickness of Metallic Coatings by the Coulometric Method

ASTM B 530: Standard Test Method for Measurement of Coating Thickness by the Magnetic Method Electrodeposited Nickel Coatings on Magnetic and Nonmagnetic Substrates

ASTM B 568: Standard Test Method for Measurement of Coating Thickness by X-ray Spectrometry

ASTM B 748: Standard Method for Measurement of Thickness of Metallic Coatings by Measurement of Cross Section with a Scanning Electron Microscope

ASTM A 754: Standard Test Method for Coating Thickness by X-ray Fluorescence

Visible contact marks resulting from electroplating operations shall be confined to the interior of the wrenching and drive openings. Surfaces of the electroplated tool shall be adherent, smooth, continuous, and free of visible defects such as blisters, pits, roughness, cracks, and uncoated areas. Surfaces shall not be stained or discolored and shall have a bright appearance. The electroplated tool shall be clean and free of damage. The tool shall be tested for adhesion of electroplated materials in accordance with ASTM B571, Standard Test Methods for Adhesion of Metallic Coatings, by one of the following methods: Grind-saw, Push, File, or Burnishing. The sample size shall be one tool per lot.

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3.3 Surface roughness. All surfaces shall be thoroughly cleaned and free from cracks, burrs, pits, nodules, apparent forge flash, and external sharp edges. All external surfaces (all surfaces excluding the wrenching surfaces and internal parts of the female drive) of the wrenches shall be fully polished to a maximum surface roughness height value of 63 micro-inches. The measurements shall be made in the surface area of the thickness of the tool. Measurements shall be made with a surface measuring instrument using a 0.03-inch roughness width cutoff. All surface roughness height values shall be rated as the arithmetic average. Nomenclature is defined in ANSI B46.1-85, Surface Texture. The sample unit shall be one tool per lot.

3.4 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.5 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

5.5 Torque test. When specified, the nut engagement of the wrench shall be fully inserted onto the hexagon mandrel secured to a testing machine. Force shall be applied to a drive handle inserted into the square drive of the

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wrench. The specified torque loads specified in Tables I and II shall be the load applied to the mandrel. The mandrel shall have a minimum hardness of 55 on the Rockwell C scale, be of sufficient length to fully engage the wrench, and conform to the following dimensions and tolerances. Hexagonal mandrel; dimensions, 9/16 inch + 0.001 inch, - 0.003 inch across flats; and 0.6356 inch minimum across corners. After application of the load test, any tool which exceeds the "C" dimension of ANSI B107.17M for the nominal size by more than 0.003 inch shall be rejected.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI B46.1-85: American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

ASTM B487, B499, B504, B530, B568, B571, B748, A754: American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) When applicable, the specific styles and forms as specified in the Commercial Item Description.
(d) If bid samples are required, the number of samples required, and the subjective inspection requirements.
(e) If first article samples are required and the number of samples required.
(f) If lot sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

(g) What preservation, packing, packaging, and marking of packages are required.
(h) What dimensions are required for style B, form B wrenches (see 3.1).

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7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the wrench specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

NSN	SPECIFICATION COMPONENT NUMBER
5120-00-277-3661	2
5120-00-277-3662	3
5120-00-277-3663	4
5120-00-277-3664	5
5120-00-277-3665	6
5120-00-222-7975	10
5120-00-288-6488	190

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS

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